PH 716 Applied Survival Analysis

Part 9: Competing risks

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Competing risks

- K (≥ 2) mutually exclusive events
  - Occurrence of one of these events precluding us from observing the other event on this subject
  - Observation for each subject terminated if
    - encountering one (and only one) of these K events or
    - censoring
  - E.g., death from different causes: natural causes, accidental death, homicide, suicide, etc.
  - E.g., disease-free survival from multiple conditions: heart disease, cancer, chronic respiratory diseases, stoke, diabetes, kidney diseases, etc.
- Notations
  - i: subject index, $i = 1, \ldots, n$
  - $\widetilde{T}_i = \min(T_i, C_i)$: observed survival time for subject i
    - T_i: authentic survival time for subject i
    - C_i: censoring time for subject i
  - Δ_i: the (re-defined) event indicator for subject i
    - $\Delta_i = k$, $k = 1, \ldots, K$: $T_i = \widetilde{T}_i$ and the event label is k
    - $\Delta_i = 0$: $T_i = C_i$
  - $x_{i1}, \ldots, x_{ip}$: values of covariates for subject i

Recall functions characterizing the survival distribution

- Limited to continuous T_i
- Hazard function
$$\lambda_{T_i}(t) = \lim_{\delta \to 0^+} \frac{\Pr(t \leq T_i < t + \delta \mid T_i \geq t)}{\delta}$$
  - The instantaneous risk of experiencing one event at time t, assuming the subject has survived up to t
- Cumulative hazard function $\Lambda_{T_i}(t) = \int_0^t \lambda_{T_i}(u)\mathrm{d}u$
- Survival function $S_{T_i}(t) = \Pr(T_i > t)$
- (Cumulative) distribution function $F_{T_i}(t) = \Pr(T_i \leq t) = 1 - S_{T_i}(t)$
- Probability density function $f_{T_i}(t) = \mathrm{d}F_{T_i}(t)/\mathrm{d}t$
- Interaction among the above functions
  - $\lambda_{T_i}(t) = -\mathrm{d}\ln S_{T_i}(t)/\mathrm{d}t = -\mathrm{d}\ln\{1 - F_{T_i}(t)\}/\mathrm{d}t$
  - $\Lambda_{T_i}(t) = -\ln S_{T_i}(t)$
  - $S_{T_i}(t) = \exp\{-\Lambda_{T_i}(t)\} = \exp\{-\int_0^t \lambda_{T_i}(u)\mathrm{d}u\}$
  - $f_{T_i}(t) = -\mathrm{d}S_{T_i}(t)/\mathrm{d}t = S_{T_i}(t)\lambda_{T_i}(t)$

Motivation to consider the event type k

- May sacrifice valuable information by ignoring the event label (i.e., merging all the K events together)
  - E.g., $\lambda_{T_i}(t) = \lambda_0(t)\exp(\sum_{j=1}^p x_{ij}\beta_j)$
    - β_1 potentially insignificant in general with x_{i1} as a strong predictor for certain specific event

Cause-specific functions characterizing the survival distribution

- Cause-specific hazard function

$$\lambda_{T_i}^{(k)}(t) = \lim_{\delta\to 0^+} \frac{\Pr(t \le T_i < t+\delta, \Delta_i = k \mid T_i \ge t)}{\delta}, \quad k = 1, \ldots, K$$

  - The instantaneous risk of experiencing event k at time t, assuming the subject has survived up to t
  - $\sum_{k=1}^K \lambda_{T_i}^{(k)}(t) = \lambda_{T_i}(t)$
- Cumulative cause-specific hazard function $\Lambda_{T_i}^{(k)}(t) = \int_0^t \lambda_{T_i}^{(k)}(u)\mathrm{d}u$
  - $\sum_{k=1}^K \Lambda_{T_i}^{(k)}(t) = \Lambda_{T_i}(t)$
- Sub-distribution function/cumulative incidence function (CIF) $F_{T_i}^{(k)}(t) = \Pr(T_i \le t, \Delta_i = k)$
  - NOT a (cumulative) distribution function
  - The probability of dying from event k up to time t, while acknowledging that the subject may die of other $K-1$ causes first
  - $\sum_{k=1}^K F_{T_i}^{(k)}(t) = F_{T_i}(t) \Rightarrow S_{T_i}(t) = 1 - \sum_{k=1}^K F_{T_i}^{(k)}(t)$
- Sub-distribution hazard function

$$\bar{\lambda}_{T_i}^{(k)}(t) = -\frac{\mathrm{d}\ln\{1 - F_{T_i}^{(k)}(t)\}}{\mathrm{d}t} = \frac{\mathrm{d}F_{T_i}^{(k)}(t)/\mathrm{d}t}{1 - F_{T_i}^{(k)}(t)}, \quad k = 1, \ldots, K$$

  - The instantaneous risk at time t of experiencing event k, assuming the subject has survived from event k up to t
  - NOT the cause-specific hazard function: $\bar{\lambda}_{T_i}^{(k)}(t) \le \lambda_{T_i}^{(k)}(t)$
- Interaction among the above functions
  - $\lambda_{T_i}^{(k)}(t) = \frac{\mathrm{d}F_{T_i}^{(k)}(t)/\mathrm{d}t}{S_{T_i}(t)}$
    - Proof: $\lambda_{T_i}^{(k)}(t) = \lim_{\delta\to 0^+} \frac{\Pr(t \le T_i < t+\delta, \Delta_i = k, T_i \ge t)}{\delta \Pr(T_i \ge t)} = \lim_{\delta\to 0^+} \frac{\Pr(t \le T_i < t+\delta, \Delta_i = k)}{\delta S_{T_i}(t)} = \frac{\mathrm{d}F_{T_i}^{(k)}(t)/\mathrm{d}t}{S_{T_i}(t)}$
  - $F_{T_i}^{(k)}(t) = \int_0^t \lambda_{T_i}^{(k)}(u) S_{T_i}(u)\mathrm{d}u = 1 - \exp\{-\int_0^t \bar{\lambda}_{T_i}^{(k)}(u)\mathrm{d}u\}$

Naive KM estimator [DM, Sec. 9.2.1]

- Assuming that
  - T_i iid across i, i.e., $T_i \overset{\text{iid}}{\sim} T$
  - T_i independent of C_i given covariates (if any)
  - Times to different events are independent (typically violated in medical cases)
    - Implying that at each time point the hazard of each event is the same for subjects at risk as for subjects that have experienced other competing events by that time
- Estimation procedure
  - Take the event k as the event of interest with other events considered as censored
  - Apply KM estimator to the resulting binary setting and then estimate the probability of survival from one event (in the absence of other causes) by $\prod_{j:t_j \le t}\{1 - \hat{\lambda}_T^{(k)}(t_j)\}$
    - $0 = t_0 < t_1 < \cdots < t_J$: unique failure times
    - $\hat{\lambda}_T^{(k)}(t_j) = d_{kj}/r_j$: an estimate of the cause-specific hazard function
      - d_{kj}: # of event k that happened exactly at time t_j
      - r_j: # of individuals at risk up to time t_j
- Underestimating the survival probability (i.e., overestimating the failure probability)
  - Potentially treating subjects that will never fail as if they could fail
  - The bias inflated when the competition when the hazards of competing events are larger

Ex. 9.1 High risk population in `asaur::prostateSurvival`

- Dataset `asaur::prostateSurvival` involves covariates as below.
  - `grade`: a factor with levels `mode` (moderately differentiated) and `poor` (poorly differentiated)
  - `stage`: a factor with levels `T1ab` (Stage T1, clinically diagnosed), `T1c` (Stage T1, diagnosed via a PSA test), and `T2` (Stage T2)
  - `ageGroup`: a factor with levels `66-69`, `70-74`, `75-79`, & `80+`
  - `survTime`: the survival time from diagnosis to death (from prostate cancer or other causes) or last date known alive
  - `status`: a censoring variable, `0` (censored), `1` (death from prostate cancer), and `2` (death from other causes)
- Consider the high risk population (i.e.,`grade="poor"`, `stage="T2"` & `ageGroup="80+"`).

```r
options(digits=4)
library(asaur)
library(survival)
sapply(asaur::prostateSurvival, class)
data.ex91 = asaur::prostateSurvival[
  asaur::prostateSurvival$grade == "poor" &
  asaur::prostateSurvival$stage == "T2" &
  asaur::prostateSurvival$ageGroup == "80+"
  ,
]
km.prost.naive = survfit(
  Surv(survTime, event=(data.ex91$status==1)) ~ 1,
  data=data.ex91
)
km.other.naive = survfit(
  Surv(survTime, event=(data.ex91$status==2)) ~ 1,
  data=data.ex91
)
plot(
  km.prost.naive$surv ~ km.prost.naive$time, type="s", ylim=c(0,1), lwd=2, col="blue",
  xlab="Months from prostate cancer diagnosis",
  ylab='Estimated survival probability',
)
lines(km.other.naive$surv ~ km.other.naive$time, type="s", col="green", lwd=2)
legend(
  "topright",
  c(
    "Prostate",
    "Other"
  ),
  col=c('blue','green'), lwd=2
)
```

KM estimator of CIF [DM, Sec. 9.2.2]

- Assuming that
  - T_i iid across i, i.e., $T_i \overset{\text{iid}}{\sim} T$
  - T_i independent of C_i given covariates (if any)
- Estimation procedure
  - Estimate overall survival $S_T(t)$ by $\widehat{S}_{T,KM}(t) = \prod_{j:t_j \le t}\{1 - \sum_{k=1}^{K} \hat{\lambda}_T^{(k)}(t_j)\}$
    - $0 = t_0 < t_1 < \cdots < t_J$: unique failure times

    - $\hat{\lambda}_T^{(k)}(t_j) = d_{kj}/r_j$: an estimate of the cause-specific hazard function
      - d_{kj}: # of event k that happened exactly at time t_j
      - r_j: # of individuals at risk up to time t_j
  - Estimate CIF $F_T^{(k)}(t)$ by $\widehat{F}_{T,KM}^{(k)}(t) = \sum_{j:t_j \le t} \hat{\lambda}_T^{(k)}(t_j)\widehat{S}_{T,KM}(t_j - 1)$

Revisit Ex. 9.1

```
options(digits=4)
library(asaur)
library(survival)
library(mstate)
sapply(asaur::prostateSurvival, class)
data.ex91 = asaur::prostateSurvival[
  asaur::prostateSurvival$grade == "poor" &
  asaur::prostateSurvival$stage == "T2" &
  asaur::prostateSurvival$ageGroup == "80+"
  ,
]
km.cif = Cuminc(
  time = data.ex91$survTime,
  status = data.ex91$status
)
km.cif

# Plot of CIFs and the overall survival function
plot(
  km.cif$CI.1 ~ km.cif$time, type="s", ylim=c(0,1), lwd=2, col="blue",
  xlab="Months from prostate cancer diagnosis",
  ylab="Probability"
)
lines(km.cif$CI.2 ~ km.cif$time, type="s", lwd=2, col="green")
lines(km.cif$Surv ~ km.cif$time, type="s", lwd=2, col="red")
legend(
  "topright",
  c(
    "CIF (prostate)",
    "CIF (other)",
    'Overall survival'
  ),
  col=c('blue','green','red'), lwd=2
)

# Stacked plot
library(ggplot2)
cuminc_data = as.data.frame(km.cif[, c('time','Surv','CI.1','CI.2')])
cuminc_data = tidyr::pivot_longer(
  cuminc_data, cols = -time, names_to = "Types", values_to = "estimate")
ggplot(data = cuminc_data, aes(x = as.numeric(time), y = estimate, fill = Types)) +
  geom_area(alpha = 0.6) +
  labs(x = "Months from prostate cancer diagnosis", y = "Probability") +
  theme_minimal()
```

Regression on cause-specific hazards

- Assuming that
  - T_i independent across i given covariates
  - The independent and non-informative censoring
  - Cause-specific proportional hazards
    - $\lambda_{T_i}^{(k)}(t) = \lambda_0^{(k)}(t)\exp(\sum_{j=1}^p x_{ij}\beta_j^{(k)})$
      - $\lambda_0^{(k)}(t)$: baseline cause-specific hazard of event k
      - $\beta_1^{(k)}, \ldots, \beta_p^{(k)}$: covariate effects varying from one event to another
    - OR $\lambda_{T_i}^{(k)}(t) = \lambda_0^{(k)}(t)\exp(\sum_{j=1}^p x_{ij}\beta_j)$, i.e., β_j shared by all the K events
- Estimation procedure
  - For $\lambda_{T_i}^{(k)}(t) = \lambda_0^{(k)}(t)\exp(\sum_{j=1}^p x_{ij}\beta_j^{(k)})$
    - Specify one event of interest and fit a Cox PH model with the remaining $K-1$ events treated as censoring
    - Repeat the above step and obtain K Cox PH models
  - For $\lambda_{T_i}^{(k)}(t) = \lambda_0^{(k)}(t)\exp(\sum_{j=1}^p x_{ij}\beta_j)$
    - First reshape the data frame in the "long format" by encoding the each row with K rows
    - Then fit a Cox PH model stratified by the encoded event lable in the long format
- When $\hat{\lambda}_{T_i}^{(k)}(t)$ is ready
  - $\widehat{S}_{T_i}(t) = \exp\{-\sum_{k=1}^K \int_0^t \hat{\lambda}_{T_i}^{(k)}(u)\mathrm{d}u\}$
  - $\widehat{F}_{T_i}^{(k)}(t) = \int_0^t \hat{\lambda}_{T_i}^{(k)}(u)\widehat{S}_{T_i}(u)\mathrm{d}u$
- Pros and cons
  - Easy to implement
  - Straightforward interpretation of regression coefficients in terms of hazard ratio BUT inconvenient to interpret coefficients in terms of contributions to CIFs
  - Bias induced since we treat other events as censoring

Ex. 9.2 Patients at "T2"-stage in `asaur::prostateSurvival`

- Consider patients with `stage="T2"`.

```
options(digits=4)
library(asaur)
library(survival)
sapply(asaur::prostateSurvival, class)
data.ex92 = asaur::prostateSurvival[
  asaur::prostateSurvival$stage == "T2"
  ,
]

# Regression on cause-specific hazards
data.ex92$status.1 = (data.ex92$status==1)
data.ex92$status.2 = (data.ex92$status==2)
cph.prost = coxph(
  Surv(survTime, status.1)~grade + ageGroup,
  data = data.ex92
)
summary(cph.prost)
cph.other = coxph(
  Surv(survTime, status.2)~grade + ageGroup,
  data = data.ex92
)
summary(cph.other)
```

```r
# Regression on cause-specific hazards with shared coefficients
## Reshape the data into the long format
data.ex92.long = NULL
K = length(unique(data.ex92$status))-1
for (i in 1:nrow(data.ex92)){
  data.curr = data.ex92[rep(i, times=K),]
  data.curr$event = c('prostate', 'other')
  data.curr$status.long=rep(0,K-1)
  if(data.ex92$status[i]>=1) {
    data.curr$status.long[which(data.curr$event==c('prostate', 'other')[data.ex92$status[i]])]=1
  }
  data.ex92.long = rbind(data.ex92.long, data.curr)
}
data.ex92.long = data.ex92.long[
  ,
  !(names(data.ex92.long) %in% c('status.1','status.2'))] # remove columns to avoid confusion
head(data.ex92)
head(data.ex92.long)

## Equivalency between two data frames
head(data.ex92)
head(data.ex92.long[data.ex92.long$event=='prostate',])
head(data.ex92.long[data.ex92.long$event=='other',])

## Cox PH model stratified by event
cph.strat = coxph(
  Surv(survTime, status.long)~grade + ageGroup+strata(event),
  data = data.ex92.long
)
summary(cph.strat)
```

Fine-Gray model

- Assuming that
  - T_i independent across i given covariates
  - The independent and non-informative censoring
  - $\bar{\lambda}_{T_i}^{(k)}(t) = \bar{\lambda}_0^{(k)}(t)\exp(\sum_{j=1}^p x_{ij}\beta_j^{(k)})$
    - $\bar{\lambda}_0^{(k)}(t)$: baseline sub-distribution hazard of event k
    - $\beta_1^{(k)}, \ldots, \beta_p^{(k)}$: covariate effects potentially varying from one event to another
- When $\hat{\bar{\lambda}}_{T_i}^{(k)}(t)$ is ready
  - $\widehat{F}_{T_i}^{(k)}(t) = 1 - \exp\{-\int_0^t \hat{\bar{\lambda}}_{T_i}^{(k)}(u)\mathrm{d}u\}$
- Pros and cons
  - Direct modeling of CIFs
  - Difficult to interpret regression coefficients in terms of hazard ratio, BUT may see the effect of covariates to CIFs directly
  - Might not satisfy the additive restriction $\sum_{k=1}^K \widehat{F}_{T_i}^{(k)}(t) \leq 1$

Revisit Ex. 9.2

- Poorly differentiated patients (`grade=poor`) have higher risk for death from both `prostate` and `other`.
- Elder patients also have higher risk for the death from both conditions.

```r
options(digits=4)
library(asaur)
data.ex92 = asaur::prostateSurvival[
  asaur::prostateSurvival$stage == "T2"
  ,
]
# Model fitting
cov1 = model.matrix(~ grade + ageGroup, data = data.ex92)[,-1]
cph.subdisthz.prost = cmprsk::crr(
  ftime = data.ex92$survTime,
  fstatus = data.ex92$status,
  cov1 = cov1,
  failcode=1
)
summary(cph.subdisthz.prost)
cph.subdisthz.other = cmprsk::crr(
  ftime = data.ex92$survTime,
  fstatus = data.ex92$status,
  cov1 = cov1,
  failcode=2
)
summary(cph.subdisthz.other)

# Predicted CIFs
cov1_new = matrix( # Specifying subjects with grade=poor and age= 70-74
  c(1, 1, 0, 0),
  byrow = T,
  ncol = 4
)
colnames(cov1_new) = colnames(cov1); cov1_new
predict.prost = predict(
  cph.subdisthz.prost,
  cov1 = cov1_new
)
head(predict.prost)

predict.other = predict(
  cph.subdisthz.other,
  cov1 = cov1_new
)
head(predict.other)

# Create overall survival
pool.time = sort(unique(c(cph.subdisthz.prost$uftime, cph.subdisthz.other$uftime)))
cifs = data.frame(
  time = pool.time,
  surv = numeric(length(pool.time)),
  cif.prost = numeric(length(pool.time)),
  cif.other = numeric(length(pool.time))
)
for (i in 1:nrow(cifs)){
  if (!(i %in% predict.prost[,1])){
    if (cifs$time[i] < predict.prost[1,1])
```

```r
      cifs$cif.prost[i] = 0
    else cifs$cif.prost[i] = cifs$cif.prost[i-1]
  }else cifs$cif.prost[i] = predict.prost[,2][predict.prost[,1] == cifs$time[i]]

  if (!(i %in% predict.other[,1])){
    if (cifs$time[i] < predict.other[1,1])
      cifs$cif.other[i] = 0
    else cifs$cif.other[i] = cifs$cif.other[i-1]
  }else cifs$cif.other[i] = predict.other[,2][predict.other[,1] == cifs$time[i]]
}
cifs$surv = 1-cifs$cif.prost-cifs$cif.other
head(cifs)

# Trajectory plot of CIFs and overall survival
plot(
  cifs$cif.prost ~ cifs$time,
  type="s", ylim=c(0,1), lwd=2, col="blue",
  xlab="Months from prostate cancer diagnosis",
  ylab='Probability',
)
lines(cifs$cif.other ~ cifs$time, type="s", col="green", lwd=2)
lines(cifs$surv ~ cifs$time, type="s", col="red", lwd=2)
legend(
  "topright",
  c(
    "Prostate",
    "Other",
    'Overall Survival'
  ),
  col=c('blue','green','red'), lwd=2
)

# Stacked plot of CIFs
cifs_long = tidyr::pivot_longer(
  cifs, cols = -time, names_to = "Types", values_to = "estimate")
library(ggplot2)
ggplot(data = cifs_long, aes(x = as.numeric(time), y = estimate, fill = Types)) +
  geom_area(alpha = 0.6) +
  labs(x = "Months from prostate cancer diagnosis", y = "Probability") +
  theme_minimal()
```